

Code/eqn - 5

Tasks - 5.

1.

$$PE = \begin{bmatrix} [PE(0,0)=5] & [PE(0,1)=6] \\ [PE(1,0)=7] & [PE(1,1)=8] \end{bmatrix}$$

2. Cycle	Input Row 0	Input Row 1	PE[0][0]	PE[0][1]	PE[1][0]	PE[1][1]
1	1	2	$1 \times 5 = 5$			
2	3	4	$3 \times 5 = 15$	$1 \times 6 = 6$	$5 + (2 \times 1) = 19$	
3				$3 \times 6 = 18$	$15 + (4 \times 1) = 43$	$6 + (2 \times 2) = 22$
4						$18 + (4 \times 2) = 50$

Final Output

$$C = [[19, 22], [43, 50]]$$

3. a. 4 outputs $\times$ 2 MAC's each = 8 MAC operations

b. No. of times each input is reused.

Each value is used 2 times in total

For A.

Used In

$A[0][0] = 1$	$C[0][0], C[0][1]$
$A[0][1] = 2$	$C[0][0], C[0][1]$
$A[1][0] = 3$	$C[1][0], C[1][1]$
$A[1][1] = 4$	$C[1][0], C[1][1]$

For B

Used In

$B[0][0] = 5$	$C[0][0], C[1][0]$
$B[0][1] = 6$	$C[0][1], C[1][1]$
$B[1][0] = 7$	$C[0][0], C[1][0]$
$B[1][1] = 8$	$C[0][1], C[1][1]$

c. Off Chip Memory Access

$$A \text{ reads} = 4$$

$$B \text{ reads} = 4$$

$$C \text{ writes} = 4$$

$$\text{Total off chip Access} = 12$$

4. In of stationary dataflow, the op values $C[i][j]$ stay fixed in the PE's until the final result is computed.